Enthalpy Change & Activation Energy

Exam code: 0620 & 0971

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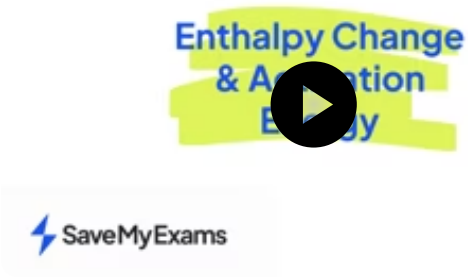

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Enthalpy change & activation energy

Extended tier only

- For atoms or particles to react with each other in a chemical system they must **collide** together
- A number of factors affect the success of a collision:
  - **Energy**
  - **Orientation**
  - **Number of collisions per second** – the **frequency** of collisions

Help me understand

What is activation energy?

- In terms of the energy of collisions, there is a **minimum amount of energy** required for a successful collision
  - A successful collision is where the particles in the reactant(s) are rearranged to form the products
- This minimum amount of energy is called the **activation energy, E_a**
- Different reactions have different activation energies, depending on the chemical identities involved
- Reactions with higher activation energies require more energy to start than those with lower activation energies

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

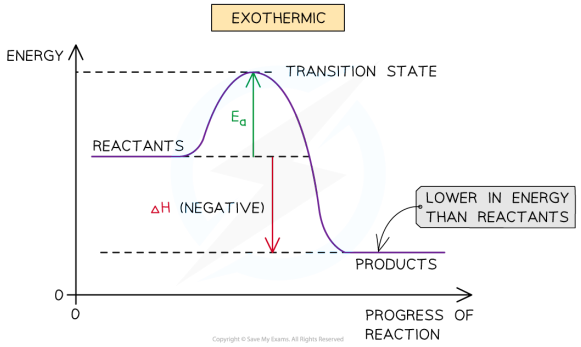
the **enthalpy change, Δ*H***, of the reaction.
- Δ*H* is:
  - Positive for an endothermic reaction
  - Negative for an exothermic reaction

Reaction pathway diagrams

Extended Tier Only

Exothermic reactions

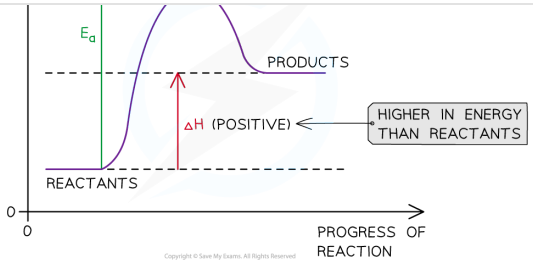
- A reaction is **exothermic** when more energy is released forming new bonds for the products than absorbed breaking the bonds in the reactants
- So, the products have less energy than the reactants
  - This means that the change in energy is negative
- Therefore, an exothermic reaction has a **negative** value for enthalpy, Δ*H*
- The reaction pathway diagram for an exothermic reaction is:



The reaction pathway diagram for exothermic reactions

Endothermic reactions

- A reaction is **endothermic** when more energy is absorbed breaking the bonds in the reactants than released forming new bonds for the products
- So, the products have more energy than the reactants
  - This means that the change in energy is positive
- Therefore, an endothermic reaction has a **positive** value for enthalpy, Δ*H*
- The reaction pathway diagram for an endothermic reaction is:



The reaction pathway diagram for endothermic reactions.

Examiner Tips and Tricks

You **must** be able to draw these pathway diagrams and label the following parts:

- Reactants
- Products
- Enthalpy change of the reaction, ΔH
- Activation energy, E_a

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